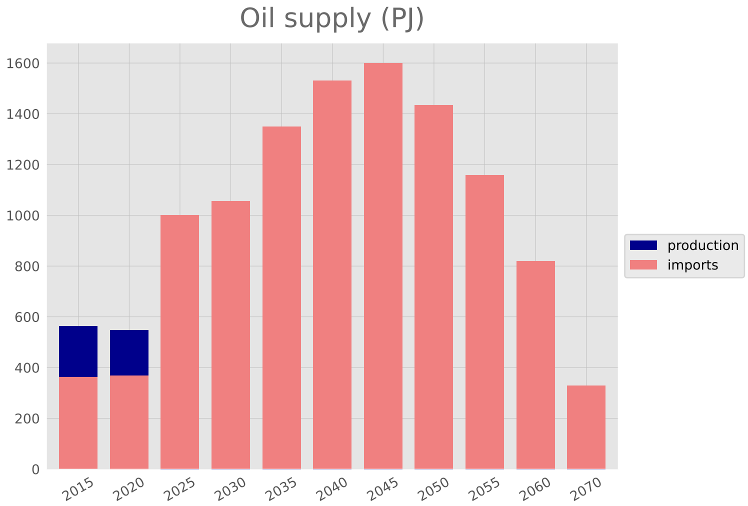
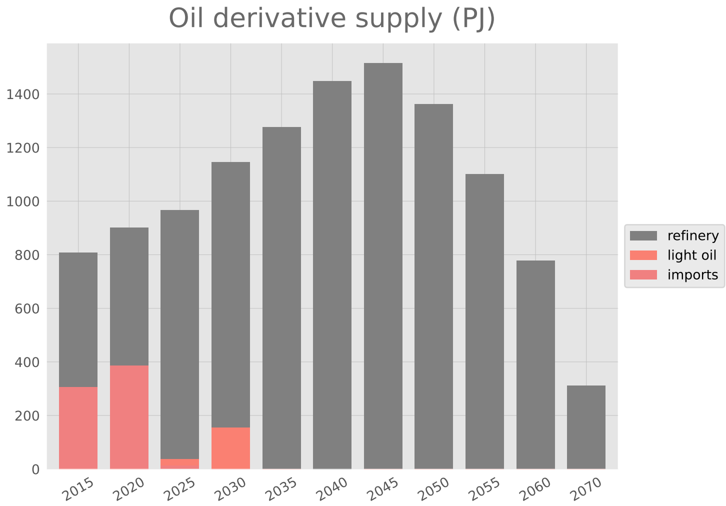
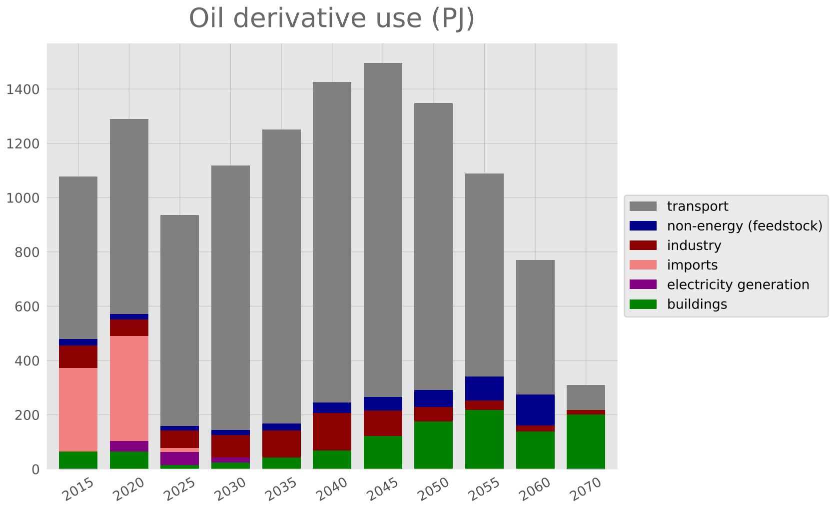
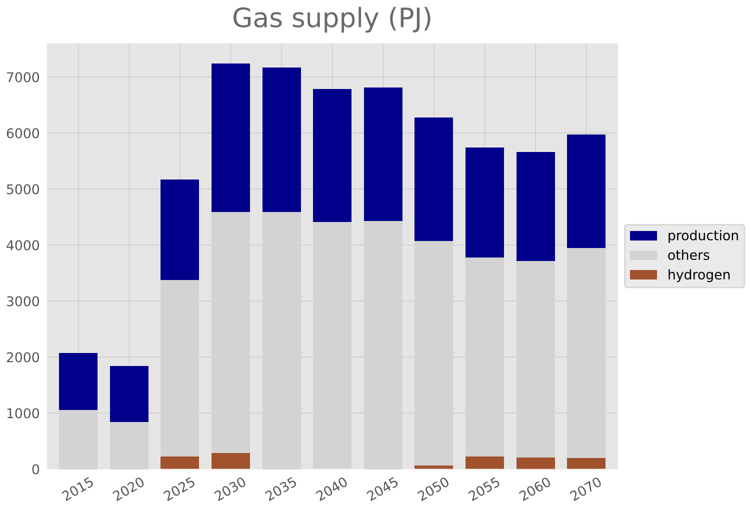


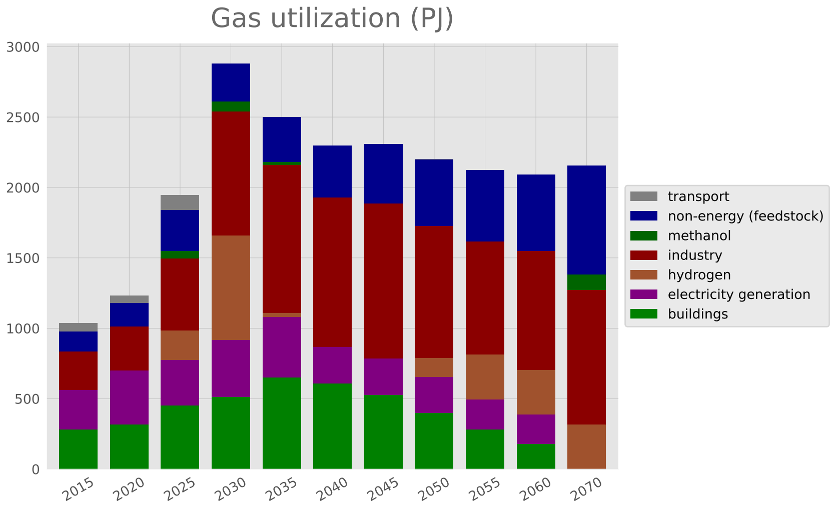
Oil supply (PJ)
1600
1400
1200
1000
800
600
400
200
0
2015
2020
2025
2030
2035
2040
2045
2050
2055
2060
2070
production
imports


Oil derivative supply (PJ)
0
200
400
600
800
1000
1200
1400
2015
2020
2025
2030
2035
2040
2045
2050
2055
2060
2070
refinery
light oil
imports

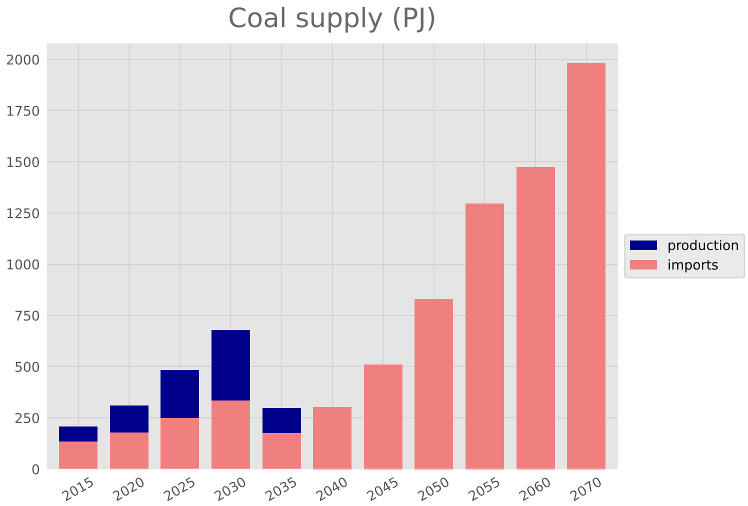

Oil derivative use (PJ)
0
200
400
600
800
1000
1200
1400
2015
2020
2025
2030
2035
2040
2045
2050
2055
2060
2070
transport
non-energy (feedstock)
industry
imports
electricity generation
buildings



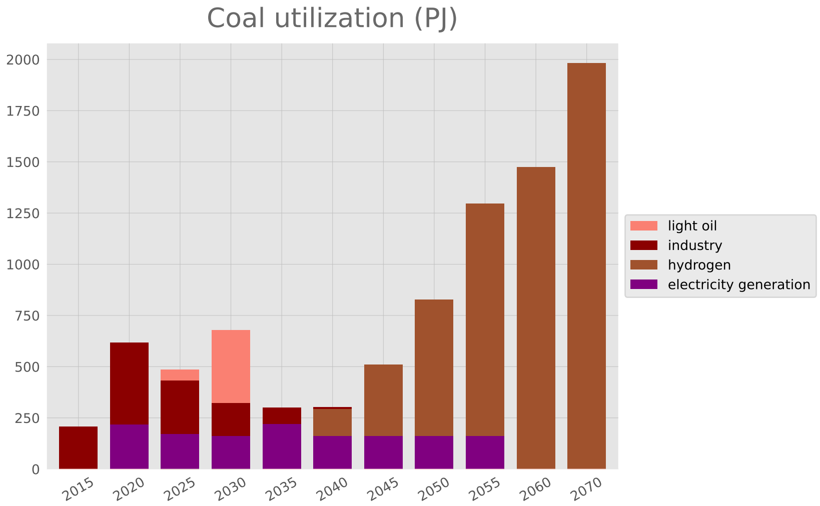
Gas supply (PJ)
0
1000
2000
3000
4000
5000
6000
7000
2015
2020
2025
2030
2035
2040
2045
2050
2055
2060
2070
production
others
hydrogen

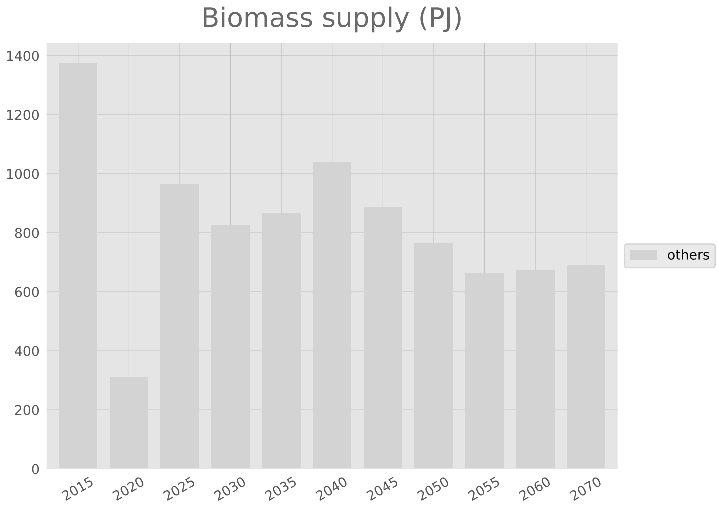


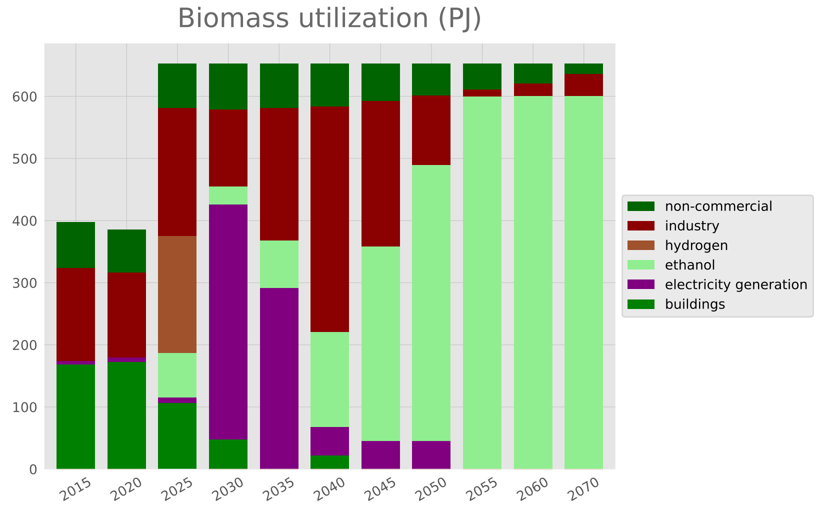
Gas utilization (PJ)
3000
2500
2000
1500
1000
500
0
2015
2020
2025
2030
2035
2040
2045
2050
2055
2060
2070
transport
non-energy (feedstock)
methanol
industry
hydrogen
electricity generation
buildings

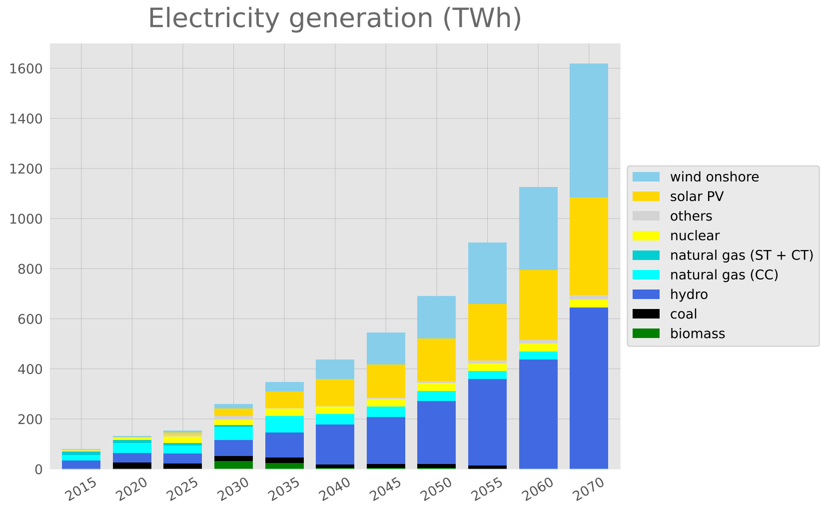


Coal supply (PJ)
2000
1750
1500
1250
1000
750
500
250
0
2015
2020
2025
2030
2035
2040
2045
2050
2055
2060
2070
production
imports

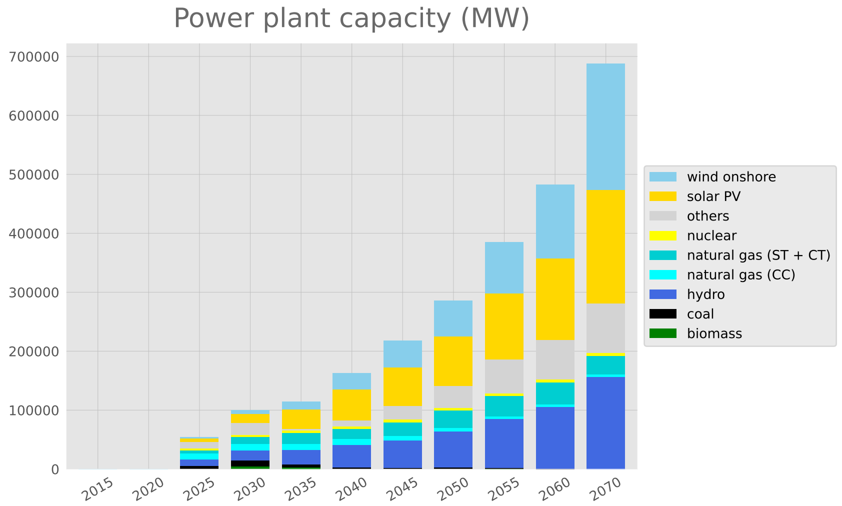

Coal utilization (PJ)
2000
1750
1500
1250
1000
750
500
250
0
2015
2020
2025
2030
2035
2040
2045
2050
2055
2060
2070
light oil
industry
hydrogen
electricity generation

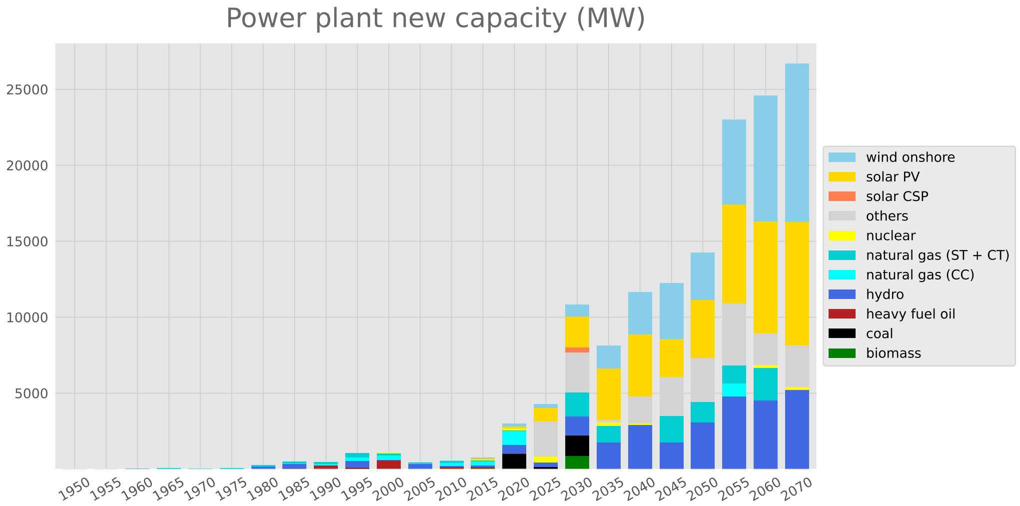

Biomass supply (PJ)
0
200
400
600
800
1000
1200
1400
2015
2020
2025
2030
2035
2040
2045
2050
2055
2060
2070
others

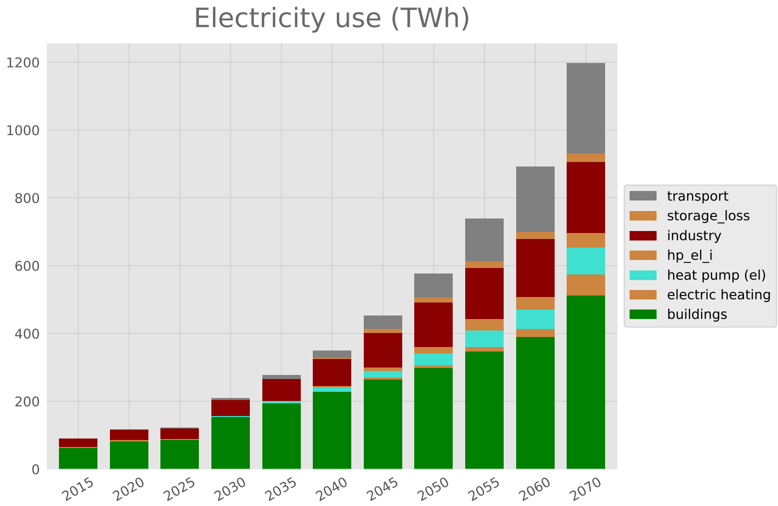

Biomass utilization (PJ)
0
100
200
300
400
500
600
2015
2020
2025
2030
2035
2040
2045
2050
2055
2060
2070
non-commercial
industry
hydrogen
ethanol
electricity generation
buildings

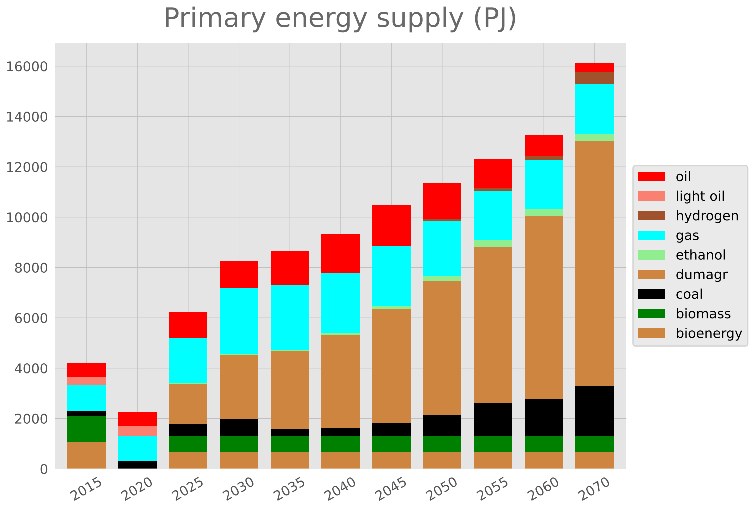


Electricity generation (TWh)
0
200
400
600
800
1000
1200
1400
1600
2015
2020
2025
2030
2035
2040
2045
2050
2055
2060
2070
wind onshore
solar PV
others
nuclear
natural gas (ST + CT)
natural gas (CC)
hydro
coal
biomass

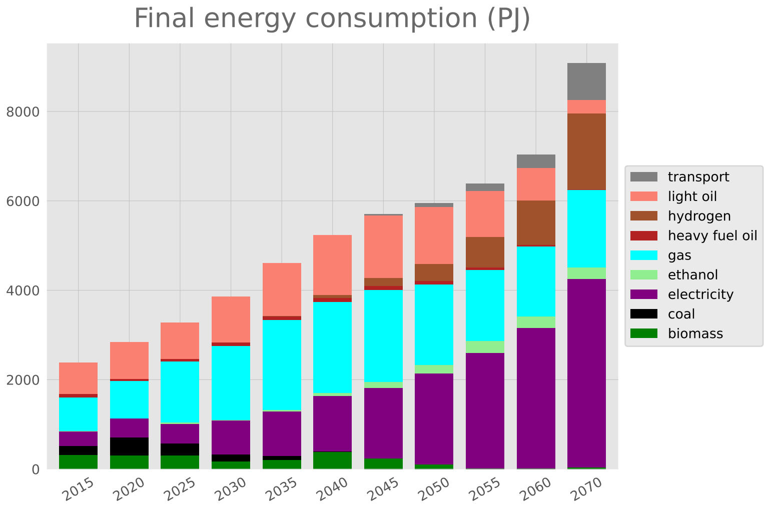

Power plant capacity (MW)
700000
600000
500000
400000
300000
200000
100000
0
2015
2020
2025
2030
2035
2040
2045
2050
2055
2060
2070
wind onshore
solar PV
others
nuclear
natural gas (ST + CT)
natural gas (CC)
hydro
coal
biomass

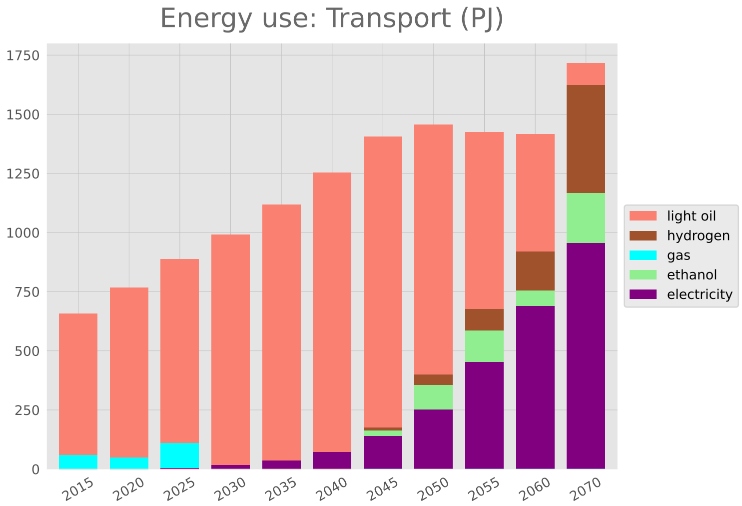

Power plant new capacity (MW)
25000
20000
15000
10000
5000
1950
1955
1960
1965
1970
1975
1980
1985
1990
1995
2000
2005
2010
2015
2020
2025
2030
2035
2040
2045
2050
2055
2060
2070
wind onshore
solar PV
solar CSP
others
nuclear
natural gas (ST + CT)
natural gas (CC)
hydro
heavy fuel oil
coal
biomass

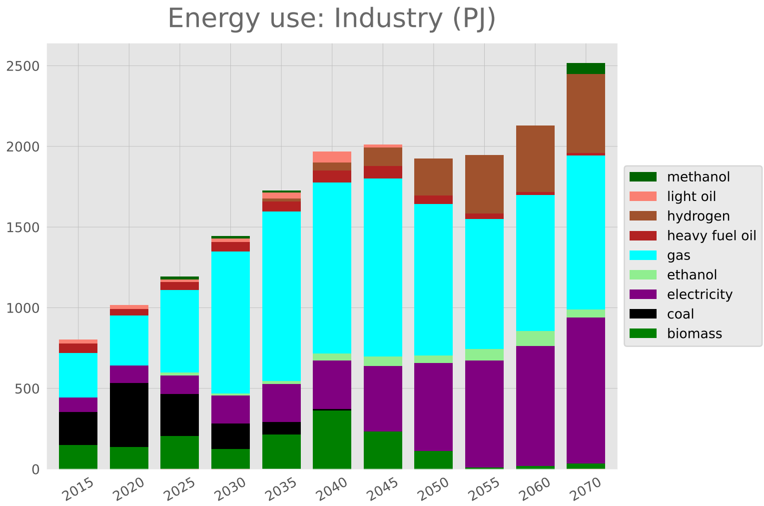

Electricity use (TWh)
0
200
400
600
800
1000
1200
2015
2020
2025
2030
2035
2040
2045
2050
2055
2060
2070
transport
storage_loss
industry
hp_el_i
heat pump (el)
electric heating
buildings



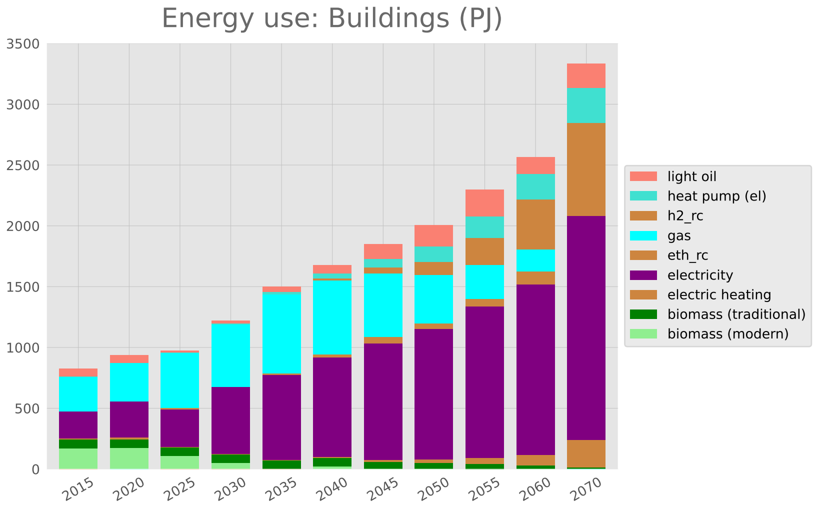
Primary energy supply (PJ)
16000
14000
12000
10000
8000
6000
4000
2000
0
2015
2020
2025
2030
2035
2040
2045
2050
2055
2060
2070
oil
light oil
hydrogen
gas
ethanol
dumagr
coal
biomass
bioenergy

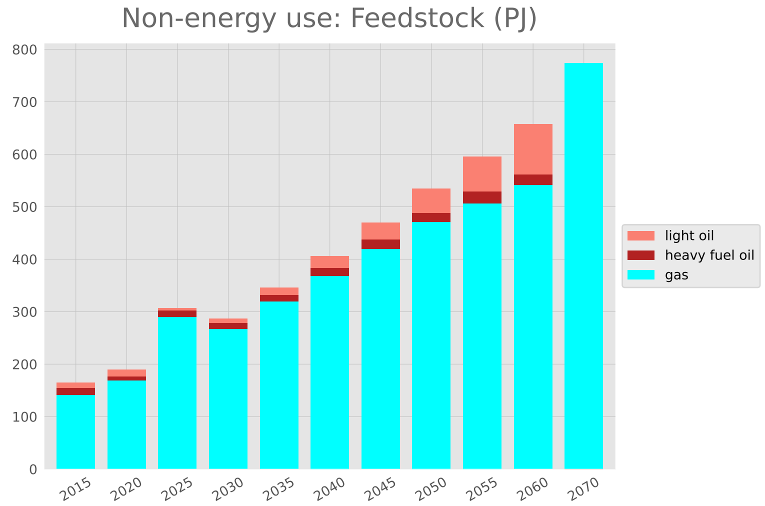

Final energy consumption (PJ)
0
2000
4000
6000
8000
2015
2020
2025
2030
2035
2040
2045
2050
2055
2060
2070
transport
light oil
hydrogen
heavy fuel oil
gas
ethanol
electricity
coal
biomass

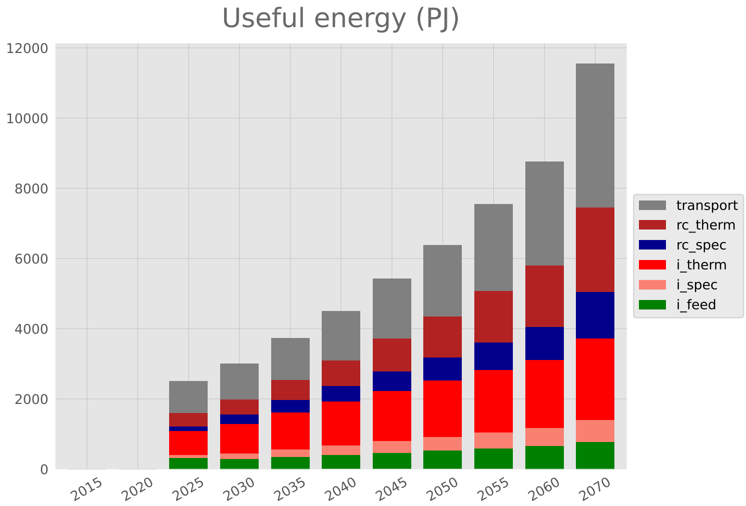

Energy use: Transport (PJ)
1750
1500
1250
1000
750
500
250
0
2015
2020
2025
2030
2035
2040
2045
2050
2055
2060
2070
light oil
hydrogen
gas
ethanol
electricity



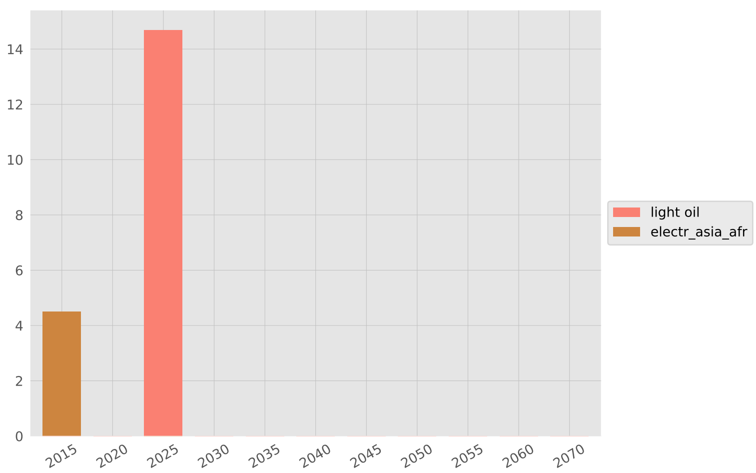
Energy use: Industry (PJ)
2500
2000
1500
1000
500
0
2015
2020
2025
2030
2035
2040
2045
2050
2055
2060
2070
methanol
light oil
hydrogen
heavy fuel oil
gas
ethanol
electricity
coal
biomass

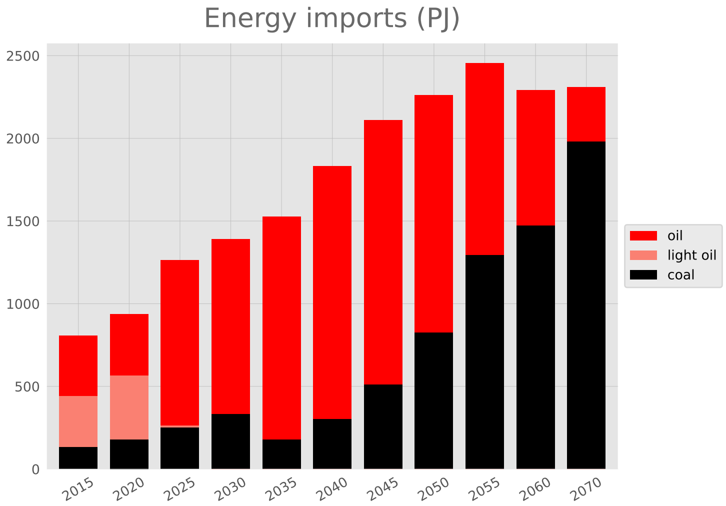

Energy use: Buildings (PJ)
3500
3000
2500
2000
1500
1000
500
0
2015
2020
2025
2030
2035
2040
2045
2050
2055
2060
2070
light oil
heat pump (el)
h2_rc
gas
eth_rc
electricity
electric heating
biomass (traditional)
biomass (modern)

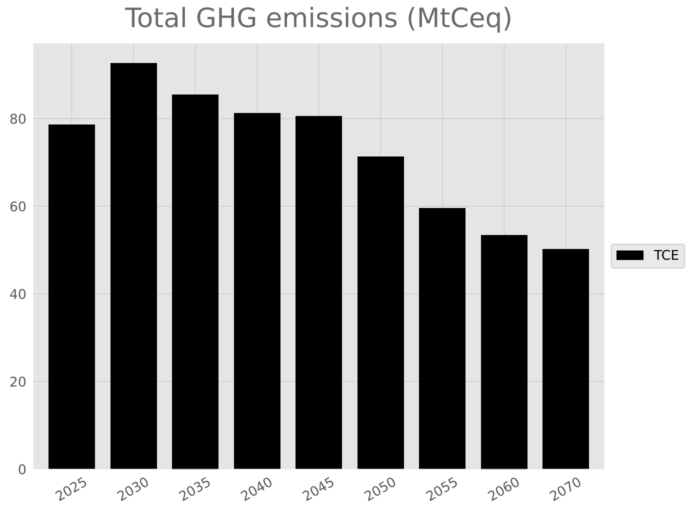

Non-energy use: Feedstock (PJ)
800
700
600
500
400
300
200
100
0
2015
2020
2025
2030
2035
2040
2045
2050
2055
2060
2070
light oil
heavy fuel oil
gas


Useful energy (PJ)
12000
10000
8000
6000
4000
2000
0
2015
2020
2025
2030
2035
2040
2045
2050
2055
2060
2070
transport
rc_therm
rc_spec
i_therm
i_spec
i_feed

Energy exports (PJ)




Energy imports (PJ)
2500
2000
1500
1000
500
0
2015
2020
2025
2030
2035
2040
2045
2050
2055
2060
2070
oil
light oil
coal



Total GHG emissions (MtCeq)
80
60
40
20
0
2025
2030
2035
2040
2045
2050
2055
2060
2070
TCE